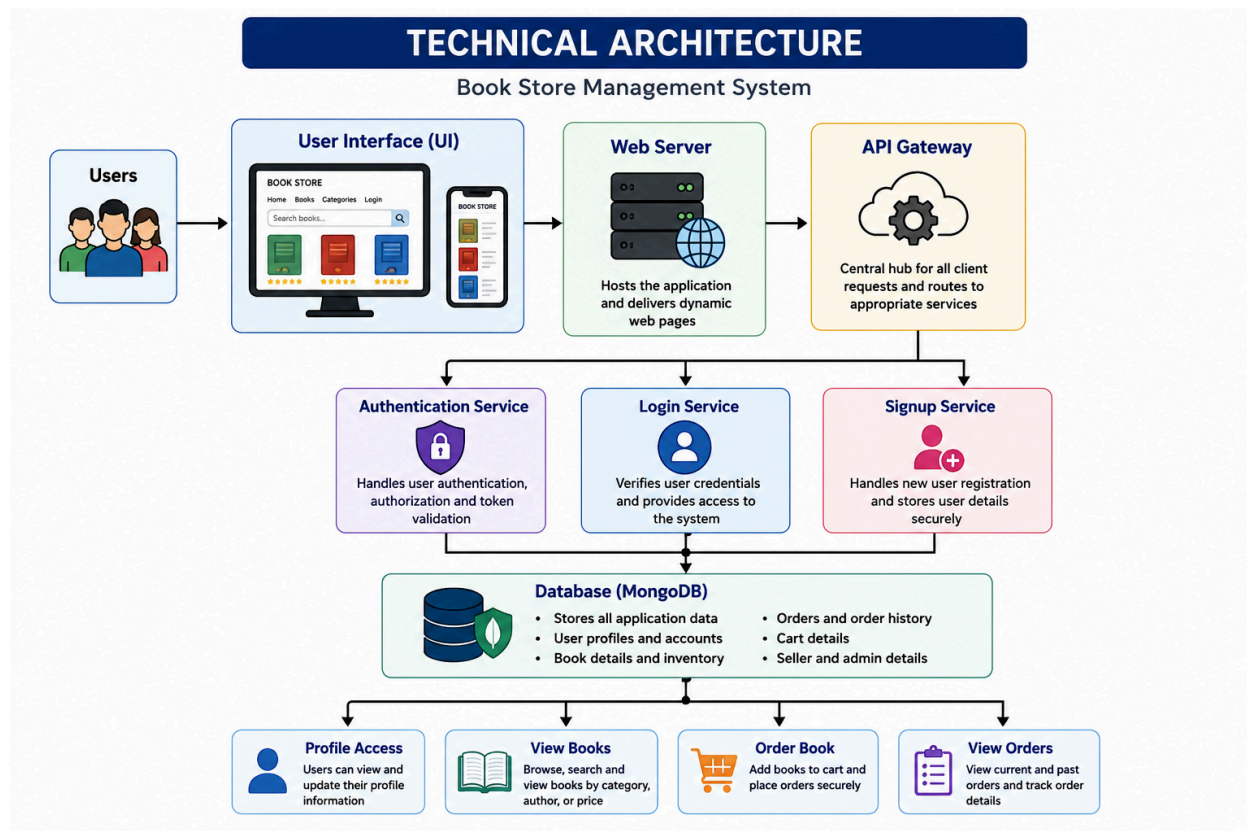


TECHNICAL ARCHITECTURE

The **Book Store application** is designed with a robust and efficient technical architecture to ensure a smooth, secure, and user-friendly reading and shopping experience for users.



1. User Interface (UI)

The front-end interface is where customers interact with the Book Store application. They can browse books, search by title or author, read descriptions, check prices, and make purchases. The interface is designed to be intuitive and user-friendly, making navigation and book discovery effortless across devices.

2. Web Server

The web server hosts the Book Store application and delivers dynamic web pages to users. It ensures responsive interactions, smooth browsing, and proper communication between the client side and the backend services.

3. API Gateway

The API Gateway acts as the central hub for all client requests. Whether the user wants to fetch book details, process orders, or manage accounts, the API Gateway routes each request to the appropriate backend service and ensures smooth communication between the components.

4. Authentication Service

This service securely handles user login, registration, and authorization. It protects sensitive user information and ensures that only authorized users can access protected features of the platform.

5. Login Service

The Login Service verifies user credentials and grants access to the application if the details are valid. It helps users access their personalized features such as profile access, browsing history, and orders.

6. Signup Service

The Signup Service handles the registration of new users by collecting user details, validating them, and storing them securely in the database.

7. Database

The database stores all important information for the Book Store application, including:

- Book details such as title, author, genre, description, price, and stock availability
- User profiles and account information
- Purchase history and order details
- Inventory and stock levels

8. Profile Access

After successful login, users can access their profile to view or update personal information, preferences, and order history.

9. View Books

This feature allows users to explore the collection of books available in the store. Users can view books by category, genre, author, or popularity.

10. Category Selection

Users can filter books by specific categories or genres such as fiction, non-fiction, science, romance, education, or children's books, making browsing easier and more personalized.

11. Order Book

This module allows users to add books to the cart, specify quantities, and complete the purchase process. It manages the ordering workflow from selection to confirmation.

12. View Orders

Users can check current and past orders, track order details, and review their purchase history.